



Genomic Medicine Programme

MPhil in Genomic Medicine – Student Handbook

ACY 2026-27



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Who is this handbook for?

The Genomic Medicine Programme at the University of Cambridge delivers the [NHS-commissioned Genomic Education framework](#) in partnership between the department of Professional and Continuing Education (PACE) and the School of Clinical Medicine (SCM).

The School of Clinical Medicine (SCM) admits full-time students onto the programme, studying for their MPhil or MRes in Genomic Medicine. The department of Professional and Continuing Education (PACE) admits part-time students onto the programme, who study Genomic Medicine alongside their work or training.

One of the key benefits of the programme is this mixing of students from many different professional and academic backgrounds.

This handbook supports and governs all full-time, MPhil students admitted onto the University of Cambridge Genomic Medicine programme by the Department of Genomic Medicine and School of Clinical Medicine (SCM).

You should follow the guidelines and adhere to the requirements set down by the Department, SCM and the wider University. Besides this handbook, further guidance will be given at various points during the course. The Genomic Medicine homepage on the Virtual Learning Environment (VLE) will contain the most up-to-date version of all documents.

If you have any questions regarding the content of this handbook, please contact your Student Administrator for the MPhil in Genomic Medicine – Mr Daniel Haddow – dh738@cam.ac.uk.

Whilst every effort is made to avoid changes to this programme, published details may be changed without notice at any time.

The Department of Professional and Continuing Education (PACE) and the School of Clinical Medicine (SCM) reserve the right to withdraw or amend any part of this programme without prior notice.

Welcome

We are pleased to welcome you to the University of Cambridge and onto the Genomic Medicine programme.

This course was originally designed in response to a request from Health Education England (HEE), the governmental body responsible for the coordination of education and training in the healthcare sector in England. They wished to promote courses designed for suitably qualified NHS healthcare professionals from across multi-professional teams in bring up-to-date knowledge of genomic techniques and practices to healthcare, and to exploit the UK's established expertise in this field.

Lectures will be co-delivered with students from all courses within the Genomic Medicine programme at Cambridge, including part-time students with Postgraduate Certificate (PGCert), Postgraduate Diploma (PGDip), Master of Studies (MSt), for credit and CPPD only modular students, as well as full-time students studying for the Master of Philosophy (MPhil) and the Master of Research (MRes). The courses attract students from many different professional and academic backgrounds. We hope that you will be able to take advantage of the different perspectives, experience, research and academic expertise that such as diverse group brings to this programme.

The University of Cambridge is one of the oldest and most prestigious academic institutions in the world. Cambridge itself is a unique place to live and study. Moreover, the Biomedical Campus is a vibrant and ever-expanding place where pioneering biomedical research takes place.

As well as the formal contents of the course, there are talks, lectures and seminars that are open to you as a member of the University. For master's level students, college life also provides a wide variety of academic, sporting and leisure activities for you to take advantage of. We encourage you to make the most of these opportunities.

We aim to ensure that the course runs as smoothly as possible, and that it is a positive experience for staff and student alike. If you do encounter any problems with the course arrangements, the taught material, or with your research project, please raise them as soon as possible, giving us the best opportunity to address them.

Enjoy your time at Cambridge, and we look forward to meeting you.

Prof Marc Tischkowitz, Programme Director & Head of Department
Professor of Medical Genetics and Genomic Medicine, Department of Medical Genetics

Dr Timothy Hearn, Course Director – MPhil/MRes in Genomic Medicine.

Learning Environment and Teaching Locations

The University of Cambridge is the UK's leading research university. In the 2021 Research Excellence Framework exercise, the University's School of Clinical Medicine (SCM) was ranked first in the UK for internationally recognised research. In partnership with associated hospitals, the University is one of only eight UK Academic Health Science Centres. This prized status is awarded by an international panel in recognition of the University of Cambridge as an internationally leading University Medical Centre, with groups that are globally competitive in translating excellence in biomedical science into effective healthcare.

Located on the southern edge of Cambridge, the Cambridge Biomedical Campus contains the [University of Cambridge School of Clinical Medicine](#) (SCM) with its various departments and associated institutions, [Addenbrookes Hospital](#), a major University Teaching Hospital with 1100 beds and regional services to the whole of the East of England, and a directly funded Research Institute – the [MRC Laboratory for Molecular Biology](#) (LMB). In addition to the LMB there are four further MRC units located on the campus.

The University's main Biological Science campus is located on the Downing Site in the centre of Cambridge and houses the [School of Biological Sciences](#), which has particular strengths in developmental, cellular and molecular research. Research within the School of Biological Sciences pursues several themes that intersect with research themes in the School of Clinical Medicine. Here we utilise the state of the art bioinformatics training facilities available in the [Craik-Marshall building](#) and the [Pembroke Teaching Rooms](#).

The [Department of Genomic Medicine](#) is based on Level 6 of the Addenbrookes Treatment Centre on the Biomedical Campus. The majority of teaching takes place in the School of Clinical Medicine building on the Biomedical Campus.

To the south of Cambridge are the [Brabham Institute](#) and the Hinxton Genome Campus with the [Wellcome Sanger Institute](#) and [EMBL-European Bioinformatics Institute](#).

This location in Cambridge of a major regional University Hospital, Medical School, and Research Institutes on a single campus, together with surrounding strengths in Biological Science within a ten-mile radius is rare within the UK and offers an exceptional environment for highest quality collaborative biomedical research, and its translation into healthcare.

Programme Aims

The educational aims of the programme are to:

- Provide professionally relevant teaching and learning informed by research in an integrated clinical and research environment,
- Develop and create a cohort of doctors and other professionals allied to medicine able to pursue and develop roles in the rapidly changing environment of Genomic Medicine,
- Prepare healthcare professionals for the adoption of genomic technologies and the increasing use of genomic information as part of the diagnostic and treatment pathway,
- Develop researchers competent in the use of genomic technologies for biomedical research,
- Develop a cohort of healthcare professionals with the ability and confidence to lead service improvement for safe and high-quality patient care,
- Develop a cohort of healthcare professionals allied to medicine with an understanding of research methodologies and clinical opportunities relevant to Genomic Medicine,
- Develop students' intellectual, practical and transferable knowledge related to Genomic Medicine,
- Encourage critical thinking related to Genomic Medicine,
- Prepare students for undertaking research degrees (PhD) in Genomic Medicine related research field.
- Provide students from the pharmaceutical, biotechnology and other industries with an understanding of the relevance of advances in genomics for current and future healthcare.

The learning outcomes related to knowledge and understanding are to:

- Demonstrate a broad understanding of key concepts in genomics
- Appraise the factors contributing to disease causality
- Critically evaluate the role of genomic technology in healthcare
- Demonstrate an understanding of the use of patient data in healthcare
- Demonstrate a comprehensive understanding of the fundamental principles of human genomics, including the structure, function and regulation of genes and genomes
- Infer modes of disease inheritance and determine the appropriate methodology to use to identify genomic causality on a case-by-case basis
- Evaluate the molecular mechanisms underlying genetic diseases and their implications for human health and disease
- Critically evaluate the role of technology in genomic research and healthcare
- Apply ethical and professional principles to the role of genomics in healthcare to reach sound decisions that consider both individual patient and societal implications
- The ability to engage in focused, professionally relevant, independent and produce a dissertation.

Learning outcomes related to skills and attributes are to develop:

- Interpret genomic variants and their clinical significance
- Apply appropriate tools to manage and interpret genomic data
- Employ advanced communication skills to convey complex genetic information clearly to, for example, patients, family members, the wider public, interprofessional colleagues and genomic science and healthcare specialists
- Design, plan and undertake an independent research project to test a hypothesis and/or critically evaluate current research in a specific area of genomic medicine
- The skills necessary to locate, read, interpret and analyse primary and secondary sources of material enabling the development of a conceptual and theoretical understanding of recent developments in genomic medicine.

Teaching Methods

Teaching takes place during week-long modules, approximately 9-5, Monday-Friday (*with the exception of GM2*). In person attendance is mandatory, it is not possible to attend modules remotely.

Teaching methods will include, but are not limited to lectures, presentations by guest speakers and facilitators, interactive learning activities, reading and assignments to be completed by students outside of classroom sessions.

- Teaching methods will include a combination of interactive seminars, large and small groups work with emphasis on peer review and reflection,
- Face-to-face delivery, with syllabuses and reading lists given to students, online support between modules and feedback on all assignments,
- Seminars, discussions and lectures led by a tutor with specialist knowledge of the content,
- Guest lectures and seminars presented by leaders in their field,
- Personal study guided by appropriate syllabuses, reading lists and resources provided,
- Full and appropriate use of the programme's Virtual Learning Environment (VLE),
- Small group in person supervisions with your programme supervisor.

Study Hours

Each module runs for one week (approximately 9-5, Monday-Friday). With the removal of breaks, each module equates to approximately 30 hours of in person teaching. It is expected that on top of the 30-hours teaching, students will attend an in-person supervision following the completion of each module, read widely around the subject matter delivered in each module and complete each module assignment.

The award of academic credit is a means of quantifying and recognising learning within the UK. One credit notionally represents 10-hours of learning. Each module is worth 15 credits, so students should expect to study for approximately 150 hours to complete each module



successfully. However, it is recognised that student's study at different paces and use a variety of approaches, so this is a recommendation, rather than a hard-and-fast calculation.

Attendance

Each module will consist of five days of teaching. Full attendance is expected, with teaching dates made available well in advance. A register for each module will be maintained and attendance recorded. If you are unable to attend during a module, due to illness or a prior appointment, you should contact your course administrator – Mr Daniel Haddow – dh738@cam.ac.uk.

Persistent non-attendance will be addressed by the Course Director.

Module Dates and Deadlines

The key dates for all modules in the 2026-27 academic year are summarised below. Detailed information, including teaching rooms and timetables, will be communicated to enrolled students prior to each module via the Virtual Learning Environment (VLE).

Whilst every effort is made to deliver teaching at the published locations as stated below, locations are occasionally subject to change.

Module	All-day teaching	In-person at	Assignment due
GM1 Fundamentals of human genetics and genomics*	12 – 16 Oct 2026	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 16 Nov 2026 @12 noon UK time
GM2 Research and statistical skills for genomic medicine	20, 23, 29 Oct 2026 5, 19, 26 Nov 2026 10 Dec 2026 (opt)	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 18 Jan 2027 @12 noon UK time
GM3 Omics technologies and their application to genomic medicine	9 – 13 Nov 2026	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 14 Dec 2026 @12 noon UK time
GM5 Bioinformatics, interpretation and data quality assurance in genomic analysis	30 Nov – 4 Dec 2026	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 11 Jan 2027 @12 noon UK time
GM6 Application of genomics in infectious disease	18 – 22 Jan 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 22 Feb 2027 @12 noon UK time
GM7 Molecular pathology of cancer and application of genomics in cancer	1 – 5 Feb 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 8 Mar 2027 @12 noon UK time
GM8 Pharmacogenomics and stratified healthcare	1 – 5 Mar 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 5 Apr 2027 @12 noon UK time
GM9 Introduction to counselling and communication skills in genomic medicine	15 – 19 Mar 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 19 Apr 2027 @12 noon UK time
GM10 Epigenetics and disorders of the epigenome	12 – 16 Apr 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 17 May 2027 @12 noon UK time
GM11 Variant interpretation	26 – 30 Apr 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Tue 1 Jun 2027 @12 noon UK time



GM12 Genomics of common and rare inherited disease	10 – 14 May 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 14 Jun 2027 @12 noon UK time
GM13 Genomics of obesity and metabolic disease	24 – 28 May 2027	School of Clinical Medicine Hills Road, Cambridge CB2 0QQ	Mo 28 Jun 2027 @12 noon UK time
MPhil Dissertation	No in-person attendance requirement. Work and meetings with supervisor(s) throughout second year of MPhil.		Proposal Fri 11 Dec 2026 Final Mon 19 July 2027 @12 noon UK time

***GM1 is the compulsory first module for all students**

University of Cambridge Term Dates

Not all taught modules run during term time; however, the University terms are relevant to several University processes and procedures.

Michaelmas Term – Tuesday 6 October 2026 to Friday 4 December 2026

Lent Term – Tuesday 19 January 2027 to Friday 19 March 2027

Easter Term – Tuesday 27 April 2027 to Friday 18 June 2027

Please also see: [Term dates | University of Cambridge](#)

Student Feedback

Your feedback is essential for the continued development and evaluation of the Genomic Medicine at the University of Cambridge. As part of your course, you will be asked to provide feedback at the end of each taught module. Please do provide your feedback as it is also important to us in developing and continuously improving the course and to your course tutors and lecturers as part of their own professional development. We aim to provide you with an update on the VLE after each module survey has closed to report back on how your feedback will be acted upon. We also ask you to complete an overall questionnaire at the end of the course. We are then able to reflect on your comments and implement changes as necessary to improve your experience and the experience of subsequent cohorts.

Academic Course Leads

Contacts for questions about course content, academic support and concerns.



Dr Timothy Hearn	Genomic Medicine Course Director – MPhil/MRes University of Cambridge School of Clinical Medicine	tjh70@cam.ac.uk
Prof Marc Tischkowitz	Programme Director Head of Department, Genomic Medicine University of Cambridge School of Clinical Medicine	mdt33@cam.ac.uk

Course Administration Team

Contact for general and technical questions, procedures, requests, advice and support. Many answers can be found in this handbook.

Mr Daniel Hadow	Genomic Medicine Student Administrator Department of Genomic Medicine	dh738@cam.ac.uk
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Module Leads

Contacts for module specific academic questions.

Dr Gemma Chandratillake	GM1 Fundamentals of human genetics and genomics	gemma.chandratillake@nhs.net
Dr Tim Hearn	GM2 Research and statistical skills for genomic medicine	tjh70@medschl.cam.ac.uk
Dr Leonardo Bottolo		lb664@cam.ac.uk
Prof Giles Yeo	GM3 Omics technologies and their application to genomic medicine	gshy2@cam.ac.uk
Dr Brian Lam		yhbl2@cam.ac.uk
Dr Tim Hearn	GM5 Bioinformatics, interpretation and data quality assurance in genomic analysis	tjh70@medschl.cam.ac.uk
Dr Kenneth Langlands		kenneth.langlands@pace.cam.ac.uk
Dr Theodore Gouliouris	GM6 Application of genomics in infectious disease	theo.gouliouris@doctors.org.uk
Dr Zachary Ardern		za6@sanger.ac.uk
Prof Marc Tischkowitz	GM7 Molecular pathology of cancer and application of genomics in	mdt33@cam.ac.uk

Prof George Vassiliou	cancer	gsv20@cam.ac.uk
Dr Emma Hodson	GM8 Pharmacogenomics and stratified healthcare	ejh49@medschl.cam.ac.uk
Prof Richard Sandford		rns13@medschl.cam.ac.uk
Dr James Whitworth		jww39@medschl.cam.ac.uk
Eleanor Davies	GM9 Introduction to counselling and communication skills in genomic medicine	eleanor.davies6@nhs.net
Isobel Turbin		isobel.turbin1@nhs.net
Prof Eamonn Maher	GM10 Epigenetics & epigenomics	erm1000@cam.ac.uk
Prof Miguel Constância		jmasmc2@medschl.cam.ac.uk
Dr Tim Hearn	GM11 Variant interpretation	tjh70@medschl.cam.ac.uk
Dr Kenneth Langlands		kenneth.langlands@pace.cam.ac.uk
Dr Katherine Schon	GM12 Genomics of common and rare inherited disease	ks339@cam.ac.uk
Dr Ramsay Bowden		arb63@cam.ac.uk
Dr Laura Fachal		lf9@sanger.ac.uk
Dr Robert Legg	GM13 Genomics of obesity and metabolic disease	rgl28@cam.ac.uk
Dr Kenneth Langlands		
Dr Timothy Hearn	MPhil Dissertation	tjh70@medschl.cam.ac.uk

MPhil – Programme Structure

The MPhil in Genomic Medicine is a 10-month taught course, administered and owned by the Department of Genomic Medicine, part of the School of Clinical Medicine (SCM).

The course is a full-time component of the University of Cambridge Genomic Medicine programme. The degree awarded is an MPhil (by advanced study). To complete the MPhil, students must complete either **eight taught module and a research project dissertation** or **ten taught modules and a literature review dissertation**.



Modules	8 modules	10 modules
Dissertation Type	Research project	Literature review

Research Project – 60 credits – approximately 10,000-12,000 words.

Literature Review – 30 credits – approximately 5,000-6,000 words.

Students must take the compulsory GM1 – Fundamentals of Human Genetics and Genomics module. Other modules are chosen from the schedule set forward by the School of Clinical Medicine in the Easter Term preceding the start of the course (see page 10+11).

Each student is assigned a programme supervisor; this is usually your Course Director. Your programme supervisor will provide appropriate academic support throughout the course. You will be invited to attend regular, small-group supervisions with your programme supervisor throughout the year.

Each student will be assigned a dissertation supervisor, once project proposals have been submitted in December. Your dissertation supervisor will be an expert in the field of your specific research. They will provide specialised support in the completion of your dissertation. It is your responsibility to arrange meetings with your dissertation supervisor throughout the duration of your dissertation.

[University Access, Email and Student Card.](#)

Student Registration Exercise

All students enrolled on an award bearing course are required to complete an annual registration exercise for the UK Higher Education Statistics Agency (HESA). This involves confirming and/or updating the personal details we hold about you. It is completed online and only takes a few minutes. At the same time, you are asked your preferred method of contact and who to contact in case of emergency.

On the final screen of the Student Registration Exercise, you be able to generate and collect your University of Cambridge username, known as your CRSid, and your initial password to log in to get your student account and email. Your email address will be in the form of [yourCRSid]@cam.ac.uk and will allow you access to Office365, including OneDrive. For more information on how to access your CRDId, see the [University Information Services](#) (UIS) website.

For more information on the Student Registration Exercise, see [Student registration | Cambridge students](#)

A link to the Student Registration Exercise will be sent to you by the Student Registry closer to the start of the course.

University Email Address

Please note that all communications relating your course will be done via your @cam email address. There will be an initial 2-week grace period at the start of your course to allow to complete the Student Registration Exercise and set up your @cam email account.

University Card

As you are studying a full-time programme you will be given a University Card. The University Card grants access to university premises such as the Central Library.

Your college will produce your University Card in the run up to the start of Michaelmas Term. Your college will contact you via your new @cam email address to arrange the pick up of your University Card from your college's porter's lodge. If you lose your card, or it expires, please contact your college.

University Account Access

Your university account – formerly known as your RAVEN account, is the University's central web authentication service. For more information click [here](#). You can collect your password on successful completion of the Student Registration Exercise.

Virtual Learning Environment (VLE)

The VLE is central to your studies on the Genomic Medicine programme. Through the VLE you will access important course information throughout your studies. You will be enrolled into the course space for each, individual module.

You will be enrolled onto the VLE using your new @cam email address. The PACE e-learning team will contact you to confirm your account and prompt you to set up a password. If you have any issues with access, or any wider questions regarding the VLE, contact your Course Administrator.

The VLE login page is here - [Log in to the site | PACE VLE](#).

You will also be enrolled into the Central University Moodle space - [Log in to the site | Moodle](#). Here you will find information regarding supervisions and university training courses.



University Policies

Policies that all students should familiarise themselves with include:

[Student Discipline](#)

[Freedom of speech](#)

[Public gatherings](#)

There are a number of legal regulations that all staff and students are obliged to comply with:

[Disclosure and Barring Service \(DBS\) checks](#)

[Copyright Compliance](#)

[Information Compliance](#)

[Data Protection Act \(including details about how we use your personal information\)](#)

[Freedom of Information Act](#)

More information on the items above can be found at:

<http://www.cambridgestudents.cam.ac.uk/new-students/rules-and-legal-compliance>

Student Welfare

The welfare of students is of the utmost importance to us at the University and there are various sources of advice and support. Your first point of contact should be your college tutor (you will be assigned a college tutor when you matriculate), your Course Director and/or your Student Administrator.

However, there may be circumstances where you wish to consult someone more independent of your daily environment. The following organisations are available:

- The [University's Accessibility and Disability Resource Centre \(ADRC\)](#)
- The [Cambridge University Students' Unions' Advice Service](#)
- [Equality, Diversity and Inclusion Service](#)

Further information on general wellbeing, mental health and graduate study can be found at www.studentwellbeing.admin.cam.ac.uk/ and www.cambridgestudents.cam.ac.uk/your-course/graduate-study

Assessment and Examination

Assessment

Each module generally requires submission of an assignment of approximately 2000-3500 words or equivalent.

Assignments are usually due approximately 4-weeks after the end of the module. Details of the assessment methodology and assignment will be provided as part of the individual taught module. For a complete overview of all modules, with content and learning outcomes, see the module descriptions at the end of this handbook. For an overview of all assessment assignment deadlines, see page 10+11.

In addition to the assignment requirements of the taught modules, MPhil students must also complete a dissertation.

There are two dissertation pathways:

- Eight taught modules and a 60-credit research project dissertation of between 10,000 and 12,000 words.
- Ten taught modules and a 30-credit literature review dissertation of between 5,000 and 6,000 words.

Rules of general application for assignments and dissertation.

- All assignments for examination should be submitted by the specified deadline and in the electronic version as specified in the assignment via the VLE.
- Candidates should not put their name on any assignment submitted for examination; they should instead use their allocated candidate number. *This does not apply to the dissertation, which is submitted as a named piece of work.*
- All coursework submitted for formal assessment and examination, including assignments and dissertation, will be checked by plagiarism detection software, Turnitin UK.

Marking Scheme for Postgraduate Awards.

Under the universal mark scheme, 60% is the pass mark and a distinction is 75% or over.

See below (page 19) for the general assignment/dissertation mark scheme.



Score	Mark Awarded	Student's work shows
Excellent		
80-100	Pass with distinction.	Evidence of the exceptional quality in relation to the criteria listed for the award of 70-79% and outstanding critical insights and thought-provoking arguments.
75-79	Pass with distinction.	An accessible, accurate and clear account. Clear assimilation and understanding of the evidence. Well informed by a wide range of relevant ideas. Excellent analyses, arguments and explanations. Exceptionally good structuring of the material with clear progression and development as the work proceeds.
Good		
70-74	Pass	An accessible, accurate and direct account. Clear assimilation and understanding of the evidence. Well informed by current ideas. Very good analyses, arguments and explanations. Very good insights and personal reflections on the material. Carefully structured and organised presentation.
Competent		
65-69	Pass	An accessible, accurate and direct account. Good analyses, arguments and explanations. Good insights and personal reflections on the material. Well-organised presentation.
60-64	Pass	An accessible, accurate and direct account. Fair analyses, arguments and explanation but with some remaining gaps or confusion. Fair degree of personal insight. Reasonably well organised presentation.
Pass threshold		
50-59	Fail	Reliance on a restricted range of evidence or irrelevant material introduced. Weaknesses of factual description. Weaknesses in the analyses, arguments and explanations. Weaknesses in the insights and reflections on the material. Weakly organised presentation with poor progression through the work.
0-49	Fail	Limited range of evidence or lack of focus. Weak understanding of the material presented. Lack of coherent argument.

Marking of Module Assignments.

Module assignments are graded independently by two assessors, “Marker 1” and “Marker 2”, according to the postgraduate marking scheme shown above (page 19). This process is anonymous and there is an internal moderation process to ensure that grades are fair. Each assessor will provide written feedback via Turnitin comments on the VLE. Students can review these comments once the marks for the assignment have been released. The final mark for the assignment is an average of the marks awarded by each assessor. Individual marker grades are not shared with each student.

Where assignments are comprised of multiple sections, different assessor pairs may mark each component and give feedback via the Turnitin comments section in the VLE. In this case final marks are an average of the individual marks awarded for subsections, unless a different weighting of subsections has been indicated on the assignment prompt.

Assignment marks are released via the VLE for all students at the same time, and students will be informed of this release via email. The Course Director will contact any students receiving a fail mark (i.e. less than 60%) to discuss the implications of this with you.

Please note that all marks are considered provisional until they have been approved by an exam board.

Marking of MPhil Dissertations

Two examiners will mark the dissertation, and each will report independently. The examiners are required to assure themselves that the dissertation is clearly written, takes account of previously published work on the subject and represents a contribution to scholarship.

In addition to the marking scheme for postgraduate awards, examiners will use a dissertation feedback form to mark dissertations. Students will be sent their dissertation feedback forms after marks have been approved by an exam board.

MPhil students can find a copy of the dissertation feedback form in the dissertation VLE space.

Resubmissions are not permitted for students in the Genomic Medicine Programme.

Moderation of Marks

Please note that all marks posted on the VLE are not considered final until they have been moderated in a meeting of internal and external examiners. Moderation takes place in September after the end of the academic year. If a student has any outstanding unmarked assignments (due to extensions) and cannot be included in the moderation meeting, this students' confirmation of their award is delayed until their grades can be moderated. This may be as late as the moderation meeting at the end of the next academic year.

Extension to Assignment Submission

Short extensions (maximum 3 weeks)

The deadline for the submission of a course assignment will always be clearly signalled to you in writing and will normally be in the course guide, published before the start of the course. All submission deadlines refer to UK Time. Check carefully if your local time differs from this.

It is important to manage your assignment deadlines, so you can submit your work on time. Your Course Director and Supervisors will be able to provide techniques on how to manage your deadlines and there are useful study tips and advice on time management in the Student Help section on the VLE. However, we understand that sometimes things happen which may affect your ability to study or prevent you from submitting your work. If this happens in the time leading up to your assignment submission you are advised to contact your Course Administration team to discuss your options.

The type of extensions you can request are:

1. Up to 7-days self-certified extension, no evidence required (one per academic year, not available for dissertation submission deadlines)
2. 7 days self-certified extension, followed by 2 weeks certified extension, evidence required.
3. Up to 3 weeks certified extension, evidence required

If you request a 7-days self-certified extension, you do not need to provide evidence to support your request nor are you required to provide a specific reason. However, it is essential that you let the course team and Supervisor/Course Director know prior to or on the submission deadline. Please be aware that this type of extension can only be requested once per academic year and is not available in relation to the dissertation.

Certified extensions will require the submission of evidence in support of your application. Please ensure you discuss your options with the course team and your tutor to fully understand the repercussions on subsequent deadlines, workload, marking or being able to complete your studies in time.

Examples of grounds for a short-term extension request:

- Short-term illness of self or dependants;
- Unanticipated changes in personal circumstances;

The following reasons would not normally be accepted as valid reasons for a short-term extension:

- Misreading information about submission deadlines;
- Holiday arrangements;
- Religious festivals;
- Social commitments;
- Technical software, hardware, network or internet problems.

To request an extension, you should email your admin team (Daniel Hadow dh738@cam.ac.uk) before your assignment deadline with information relating to which extension you are requesting.

Regardless of the nature of your extension request, for certified extension requests you will normally be asked to provide documentary evidence to support your application. The exact evidence will depend on the nature of the claim but may include a medical letter from your GP or hospital, a letter from your employer or legal documents. Any evidence provided must be time relevant to the period you are making a claim for. If the supporting documentation is not in English, a translation must accompany the document. All supporting documentation must be submitted together with the request for an extension, or the request will be rejected.

All documentation will be treated confidentially within the department and will only be viewed by those necessary in the decision-making process. If an extension is granted, you would be expected to meet the agreed deadline.

Please be aware when requesting an extension of the repercussions on subsequent deadlines. Falling behind due to an extension is not a valid reason for requesting another extensions to subsequent assignments. Please also be aware that extensions may result in not completing with your cohort and delays in receiving ratified marks.

Applications to extend the end of registration date should be made online via the CamSIS self-service: <https://www.cambridgestudents.cam.ac.uk/your-course/postgraduate-study/your-student-status/extending-your-submission-date>

Please note that extensions to your dissertation submission deadline may affect the timeliness of your marks being approved and subsequently when you will receive your award. It is likely that in the event you receive a longer extension to your dissertation deadline you will not receive your award and therefore graduate until the next academic year. This will have a knock-on effect on your ability to take up a place on another programme, for example a PhD, or access a graduate visa to stay in the UK if you are an overseas student.

Please note that it is not possible to retrospectively apply extensions after the original submission date.

Students with University SSDs (Student Support Documents)

If you have (or think you may have) a disability or other longstanding circumstances that will *always* impact your ability to submit assignments on time, please contact the University's [Accessibility & Disability Resource Centre](#) at the beginning of the academic year. They will work with you to assess if they can issue a University SSD. An SSD may include instructions about modifications to marking and submission deadlines. If the Genomic Medicine course administration team receives an SSD for you, they will contact you to discuss any specifications around assessment. Students with SSDs may receive a "default" extension of two weeks to every assignment. Students with such arrangements may still follow the extension procedures

outlined above if they require additional time for an individual assignment due to unforeseen circumstances.

Presentation of Assignments

Word Count

It is not a requirement to write up to the word limit for your assignment. It is provided as an upper limit but also provides guidance on the number of words expected to complete an assignment thoroughly. Whilst some individuals can write in a very concise manner, you might consider whether you are missing any aspects of the assignment if your word count is much lower than the assignment limit. Figure legends, tables, references, and the question itself should not be included in the word count. The word count provided on submission should accurately reflect the word count of the submitted assignment.

Referencing

A reference or citation is a way of acknowledging where you made use of the work of others and the correct presentation of citations and references is an important part of any piece of academic writing. Students are expected to cite and reference correctly to avoid committing academic misconduct.

To reference and cite correctly, a recognised reference system should be used. Please see the University's page on recognised reference systems: [Referencing conventions | Plagiarism and Academic Misconduct](#)

Students on the Genomic Medicine programme can use either the [Harvard style](#) or [Vancouver style](#) in all their work for the course.

The use of a citation management tool such as EndNote, Mendeley or Zotero is strongly recommended.

Genetic Nomenclature

Human gene names should be capitalized and italicized e.g. *TP53*. A gene has the same name whether it includes a variant/mutation or not. Everyone/every cell has the gene (in most cases two copies of it), but not everyone/every cell has the variant/mutation. So, the phrase “driver mutations like *TP53*” is incorrect as *TP53* is a gene not a mutation/variant. It should be phrased “driver mutations like mutations in *TP53*”, similarly “She has *BRCA1*” should be phrased “She has a pathogenic variant in *BRCA1*” since we all have *BRCA1*. Scientists will often refer to a change in a gene as a mutation, but clinically, the term “variant” is preferable, both because the pathogenicity of a variant is often not clear and the term variant carries fewer negative connotations. This site is useful if you are trying to understand variant nomenclature:

<http://varnomen.hgvs.org/bg-material/basics/>

Figures and Tables

Figures and tables can often help make a point more clearly and careful selection of figures/tables and/or drawing of original figures/tables can demonstrate your understanding of the material. Guidelines for the inclusion of figures/tables:

- If including a figure/table from the literature, either through direct copy & pasting or by redrawing a figure heavily inspired by a figure/table from the literature, such figure/tables should be properly referenced.
- If drawing a completely original figure/table, please indicate that this is your own work since it can be difficult to distinguish original figures from un-referenced figures and we want to ensure you receive appropriate credit for your work.
- Figure legends/table text should not be included in word counts.
- Free text should not be turned into a figure/table simply to extend word count

Presentation of the MPhil Dissertation

The areas which are considered in the presentation and marking of the dissertation include: presentation of assignment; clarity of expression and logical flow; referencing; discussion of literature; use of bioinformatics theory; critical approach; appropriate use of statistics and interpretation of data; and use of figures / tables. The dissertation should contain specific sections: Background, Methods, Results, Discussion, and Conclusion.

More detailed guidance on the writing and assembly of the dissertation is contained in a separate booklet that will be provided on the dissertation VLE space.

Good Academic Practice

University guidance on good academic practice and plagiarism, including information on study skills and referencing conventions: <https://www.plagiarism.admin.cam.ac.uk/>

Turnitin

All assignments and dissertations will be submitted to Turnitin UK text-matching software. Turnitin compares the text of submitted work to sources in its database, which is made up of internet content, selected journals, and previous student submissions. Turnitin is only one method of checking the originality of a student's work and examiners may use other investigative procedures if they have any queries regarding originality. See <https://www.plagiarism.admin.cam.ac.uk/investigating/turnitin>

Academic Misconduct

The University defines academic misconduct as the “gaining or attempting to gain, or helping others to gain or attempt to gain, an unfair academic advantage in formal University assessment, or any activity likely to undermine the integrity essential to scholarship and research. It includes being in possession of unauthorised materials or electronic devices during an examination, including recording or communication devices or devices that can store data, even where registered students are unaware that such materials or devices are unauthorised, have no intention of using them, or are unaware that they have them in their possession.”

Academic misconduct includes:

- **Plagiarism:** using someone else’s ideas, words, data, or other material produced by them without acknowledgement.
- **Self-plagiarism:** using the registered student’s own ideas, words, data or other material produced by them and submitted for formal assessment at this University or another institution, or for publication elsewhere, without acknowledgement, unless expressly permitted by the assessment.
- **Contract cheating:** contracting a third party to provide work, which is then used or submitted as part of a formal assessment as though it is the registered student’s own work.
- **Collusion:** working with others and using the ideas or words of this joint work without acknowledgment, as though it is the registered student’s own work, or allowing others to use the ideas or words of joint work without acknowledgment.
- **Impersonating someone or being impersonated** in an examination or arranging for someone to impersonate someone else by sitting their examination.
- **Fabrication, falsification or misrepresentation** of data, results or other outputs or aspects of research, including documentation and participant consent, or presenting or recording such data, etc, as if they were real; or
- **Failure to meet legal, ethical and professional obligations** in carrying out research. This includes failure to follow agreed protocol if this failure results in unreasonable risk or harm to humans, other sentient beings or the environment, and facilitating misconduct in research by collusion in, or concealment of, such actions by others. It includes any plan or conspiracy to attempt to do any of these things.

Please ensure you are familiar with the University’s Plagiarism and Academic Misconduct guidance for referencing:

<https://www.plagiarism.admin.cam.ac.uk/resources-and-support/referencing/referencing-conventions>

Penalties

Penalties for late submission

The following penalties apply to all individual module assignments and your final dissertation.

Where a student does not submit their assignment by midday UK time on the submission deadline day, or where an extension to the submission deadline has not been granted, the following penalties shall apply:

Penalty of 5% deducted from the score for the piece applied for each day, or part thereof, after the deadline date. Up to a maximum of 5 days. Any late penalties incurred up to 5-days will not reduce your mark to below the pass threshold (60%).

More than 5 whole days after the deadline would result in a Fail mark being applied to that piece of work.

Penalties for word count violations

The following penalties apply to all individual module assignments and your final dissertation.

It is not required to write up to the upper word limit for your assignment/dissertation. It is provided as a guide on the number of words expected to complete an assignment thoroughly. Whilst some individuals can write in a very concise manner, you might consider whether you are missing any aspects of the assignment if your word count is much lower than the assignment range. The word count should not include appendices, bibliography, tables or figures. You will include the word count on your submission and checks will be carried out by the Course Administration Team to ensure that the declared word count is accurate.

The word count **should** include in-text citations. Students may consider using the Vancouver referencing system which minimises in-text word counts – see referencing on page 23.

The word count **should not** include appendices, bibliography, tables, figures or figure legends. However, adding long narratives into tables, figures or legends **should not** be used as a means to evade word count penalties.

Should the word count go above the upper limit allowed for the assignment/dissertation you will be contacted and asked to submit an assignment/dissertation that complies with the word limit. If this results in submitting after the deadline (and permission has not been given for an extension to the deadline) then late submission penalties apply as per those set out in 'penalties for late submission'.

Should you not resubmit your assignment to comply with the word count, the assignment will be marked as a fail.

Resubmitted assignments must be resubmitted within the word count within 5-days of the original deadline (if an extension has not been granted), failure to do so will result in the assignment being marked as a fail.

Resubmissions

Resubmissions are not permitted on an assignment or dissertation in the case of a fail mark. Instead, students are able to apply to the EAMC at the end of their course (see mitigating circumstances below).

Mitigating Circumstances

The mitigating circumstances process is intended to help those who have experienced severe or unexpected difficulties at the time of submitting an assignment. The effect on the submission or performance of the assignment must have been seriously adverse, unanticipated and beyond your control.

This process is applicable **only after the deadline** for the assignment submission. If you are anticipating difficulty in advance of an assignment deadline, you should contact your student administrator to discuss available extension options. For the majority of cases, you will need to submit evidence with a mitigating circumstances request.

For more information relating to mitigating circumstances, click here - [About the Examination Access and Mitigation Committee \(EAMC\) | Cambridge students](#)

To apply for mitigating circumstances, click here - [Guidance notes and application forms | Cambridge students](#)

Use of AI

Please ensure you are familiar with the following University policies and guidance documents regarding the use of AI:

[Plagiarism and Academic Misconduct: Student Responsibilities](#)

[Plagiarism and Academic Misconduct: Artificial Intelligence](#)

[Guidance: Using Generative AI](#)

[Generative AI and Assessment | Blended Learning Service](#)

Intellectual Property (IP)

The University's IP policy allows it to fulfil its duty of care to employees and students by requiring disclosure, setting conditions for ownership of intellectual property arising from collaborative environments, and specifying a dispute resolution procedure, to which all parties within the University have access.

The full text of the IP policy is to be found in Chapter XIII, from page 1079 of Statutes and Ordinances of the University of Cambridge:

<https://www.admin.cam.ac.uk/univ/so/pdfs/2022/Cambridge-Statutes-and-Ordinances-2022.pdf>

Students contemplating research projects including data from the 100,000 Genome Project should be aware that separate arrangements may apply regarding IP generated during such projects. More information will be made available when projects are selected and during the information governance training received for accessing the 100,000 genomes project data.

General Research Ethics

The University of Cambridge has a long-standing commitment to the highest standards of ethics in research. Over time, discipline-specific policies and processes have developed to ensure that research at Cambridge not only meets the requirements of legislation and research sponsors, but also satisfies the standards of disciplinary best practice adopted by peers. See [Research Ethics | Research Integrity](#)

For information regarding General Data Protection Regulations (GDPR), see www.information-compliance.admin.cam.ac.uk/data-protection/general-data-protection-regulation and www.information-compliance.admin.cam.ac.uk/data-protection/research-participant-data/.

Research work requiring ethics approval will not be required for module assignments. However, if you have any questions or concerns, please contact your Course Director.

Notification of Results and Awards

This section describes how and when students are informed if they officially passed their course of study and will receive their award.

Notification of Results

In September at the end of the annual examining and moderation processes the Degree Committee of the School of Clinical Medicine and Veterinary Medicine, as the awarding body, notifies MPhil students of their award.

Awards and Transcripts

Until the moderation of marks and notification of results, as described above, are complete, results are not considered final. However, students may at any time during their studies request a provisional transcript from the Genomic Medicine course administration team.

Following the completion of the moderation of marks and notification of results, as described above, MPhil students will be able to see their final marks in CamSIS and will be able to download unofficial transcripts.

MPhil students will receive their final printed degree and official transcript through their college – see the section on graduation on page 31.

Additional Information for MPhil Students

Supervisions

Part of the University of Cambridge provision for matriculated students is the assignment of an MPhil supervisor and a dissertation supervisor. Early contact with your supervisors is essential, and you should keep in regular touch. You are expected to:

- Be proactive in arranging your supervisions.
- Attend and be punctual for supervision.
- Present any written work in good time.
- Raise any problems of any kind with your supervisor who will be able to advise you further.
- Understand that supervision will involve serious discussion of the progress of your course work and research.

MPhil Supervision

During your MPhil you will be allocated a supervisor. Generally, your supervisor will meet you to discuss your progress, provide feedback on assessments and review a draft of dissertation/literature reviews. Your supervisor will likely be the Course Director.

Your MPhil supervisor is also responsible for submitting termly supervision reports that serve as a record of your progress and identify any areas for improvement. You will be notified that a report has been submitted, and you are strongly encouraged to comment on it. Each Michaelmas term, you are invited to complete a self-evaluation report instead of a termly supervision report. The self-evaluation report process is student-led and is an opportunity for you to reflect on your progress to date and highlight any concerns and support needs.

MPhil Dissertation Supervision

In addition to the MPhil supervisor outlined above, all MPhil students will have a dissertation supervisor, strongly aligned with your research goals. You will be invited to a meeting with your Course Director to begin a dialogue at an early stage in Michaelmas term in regard of this. A

local supervisor will likely be already working with the data you are analysing or already leading an ongoing research project that you will join.

It is anticipated that they will support you because you will contribute to their research. It is expected that your dissertation supervisor will help you clarify your own ideas and guide your own independent efforts. They will advise on the scope and structure of your dissertation and will help you plan your work.

Your relationship with your dissertation supervisor is a very important part of your course. Your supervisor advises you in relation to your dissertation and engages you in a critical dialogue while you think, research, draft and write. Your dissertation supervisor is responsible for ensuring that you are aware of health and safety issues pertinent to your dissertation work and will advise you about any additional ethical or legal considerations regarding your research.

If you are unable to contact your dissertation supervisor, or do not receive sufficient guidance or encounter any other serious difficulty, you should report this to your MPhil supervisor or Course Director in confidence. If you feel unable to approach them, please contact your college tutor or your course administration team.

MPhil Dissertation

The dissertation is a substantial element of the MPhil course and its examination. We recommend that you start to think about your dissertation as early as possible. You should begin to formulate your ideas during Michaelmas Term and have discussions with your peers; module leads and your Course Director.

Your final dissertation title must be agreed with your supervisor and then approved by the Course Director on behalf of the Degree Sub Committee. Sometimes, changes to a particular title may be requested, wherever possible consulting the student and supervisor concerned. Once approved, titles cannot be changed without the approval of the Course Director.

The deadline for submitting your final dissertation project proposal is **Friday 11 December 2026**.

The deadline for submitting your final dissertation is **Monday 19 July 2027**

Detailed dissertation guidance will be provided in the dissertation VLE, including information about style and layout, academic writing, referencing and scholarly conventions, and research ethics.

College Induction

Colleges may expect or require you to attend induction events. Your college will send you information about induction events and matriculation nearer the time of the course starts. Please note that matriculation is **compulsory**. You cannot be awarded your degree if you have not matriculated. General information about inductions and matriculation can be found at <https://www.cambridgestudents.cam.ac.uk/new-students/arrival>.

College Tutors

The thirty-one colleges have agreed their own guide for graduate students, which is available at: <http://www.graduate.study.cam.ac.uk/colleges>

This sets out the common core of provision that all Cambridge colleges make for their graduate students, including pastoral support. It explains how this provision works in the college setting, indicating at the same time the diversity that is built into the system. The document also sets out the responsibilities of graduates as members of colleges. College graduate tutors are available for pastoral support and can help with accessing hardship funds. Each college has separate systems in place for graduate tutors, these will be made aware to you by your college during your first weeks in Cambridge.

The Students' Unions' Advice Service also offers free, confidential and independent support to all Cambridge University students, and employs two professional advisors. Contact and service details are available at <http://www.studentadvice.cam.ac.uk>.

Graduation

Following notification of your award by The Degree Committee of the School of Clinical Medicine in September, at the end of the academic year, your college will arrange your graduation ceremony or congregation (as it is formally known). You should consult the Graduate Secretary and/or Praelector of your college directly for further information.

For a detailed description of degree ceremonies at the University of Cambridge, see: www.cambridgestudents.cam.ac.uk/your-course/graduation-and-what-next/degree-ceremonies

Student Concerns and Complaints

We hope that your time studying with will be fulfilling and rewarding, however, should you have any issues or concerns about your course or study we encourage you to talk to the course director, course administration team, and/or your college tutor at the earliest opportunity to ensure the matter can be dealt with promptly. The University also has a number of procedures in place to bring these matters to a productive resolution.

Appeals, Complaints and Examination Reviews

The University's Office of Student Conduct, Complaints & Appeals handles appeals and complaints relating to all registered students, including those on MPhil degrees. Information on the University's appeals and complaints procedures, including deadlines for submission, for informal and formal academic complaints and, exam reviews is available at www.studentcomplaints.admin.cam.ac.uk/

The Office for Student Conduct, Complaints and Appeals (OSCCA)

OSCCA is part of the central University and provides procedural information, case handling and oversight of a number of formal student procedures, including student complaints, examination results reviews, student discipline and fitness to study. It is the point of contact for the external ombudsman, the Office of the Independent Adjudicator (OIA), with whom students can raise complaints following the completion of an internal University procedure. Further information at www.studentcomplaints.admin.cam.ac.uk/

Systems and IT

		Need help?
VLE for all students (used for teaching materials)	https://vle.pace.cam.ac.uk/login/	elearning@pace.cam.ac.uk or 'eLearning Helpdesk' link in the VLE
Moodle for graduate students (used for department information and university training)	https://www.vle.cam.ac.uk/	moodlehelp@uis.cam.ac.uk
CamSIS	https://camsis.cam.ac.uk	servicedesk@uis.cam.ac.uk
University IT Helpdesk for all students	https://help.uis.cam.ac.uk/	servicedesk@uis.cam.ac.uk

Virtual Learning Environment (VLE)

The Genomic Medicine programme is supported and delivered in part through an online VLE, which is central to the course, and a vital study resource and communication channel. It is also the place where assignments are submitted, and feedback is received.

The course material, assessment and timetables are all delivered over the VLE. Presentations given by lecturers and speakers will be uploaded to the VLE. Not every speaker is willing to share their lecture material, and some may contain unpublished or proprietary information. It is strongly suggested that you take extensive notes unless you are advised otherwise.

On the VLE, you will also have access to many other online resources to enable you to get the most from your studies.

If you encounter any problems, please contact the VLE Helpdesk: elearning@pace.cam.ac.uk

Getting Started – Links, Guides and Training for IT and Computing

For IT guides and useful links to further information this is the best place to start:

<https://help.uis.cam.ac.uk/new-starters/it-for-students>

There are links for setting up Eduroam internet access and many more IT related items, including information about the various passwords and how to collect and set them up.

The University IT support service desk is at <https://help.uis.cam.ac.uk/>

Tel: 01223 762999 Email: service-desk@uis.cam.ac.uk

Students are expected to take personal responsibility for ensuring that they are familiar with the software packages that they will need during the course of their studies. The University Computing Service offers a wide variety of instructor-led and self-taught training courses for beginners and advanced users at <https://www.training.cam.ac.uk/ucs/>

Inexpensive laser printing, photocopying and other services are also available at the Cambridge Student Union, see: <https://www.cambridgesu.co.uk/aboutus/services/printshop/>
MSt students may also be able to access printing and IT support through their colleges.

Please ensure you back up your course files and in particular assignment and dissertation files, as technological problems do not constitute good grounds for seeking extensions to assignments or your dissertation.

Software

The University's Information Services has negotiated a number of arrangements which provide free or heavily discounted software for Cambridge students, see: <https://www.uis.cam.ac.uk/>

Office 365 is available free for students.

Wi-Fi

Wi-fi facilities - UniofCam and Eduroam - are available throughout the University. You will require a University Account password to gain access to UniofCam and a 16-digit code for Eduroam. Eduroam provides internet access in a wide and increasing number of universities around the world.

University of Cambridge Resources

Students have access to a wide range of facilities, discounts, training, and services to ensure your time with us is as enjoyable and productive as possible. Some of these include:

Talks, Seminars and Conferences

There are many talks and seminars that may be particularly relevant to students of Genomic Medicine.

These seminars provide a wide perspective on clinical research and practice and are usually delivered by outstanding speakers. The following series may be of particular interest:

[Clinical School Seminar Series](#)

[CRUK seminar series](#)

Genomic Medicine Newsletter

Students on the Genomic Medicine programme are subscribed to a weekly course newsletter which will highlight course news as well as many additional opportunities for Genomic Medicine specific learning and engagement.

Cambridge University Libraries

There are numerous university libraries which you can access, some requiring the University card. The University Library (on West Road), the Medical Library (in the Clinical School building) and your own college library will be particularly useful. The Medical Library in the Clinical School affords a wide spectrum of books and journals.

The University Library

The University Library (UL) on the outskirts of the city centre is likely to be your main source for research and is affiliated with over 20 University departmental and faculty libraries in Cambridge. It is a legal deposit library and a repository of official documents of the UK government and of many international organisations. To gain access to the library, you will need to have your University Card as this also serves as your University Library card and is a prerequisite for registration at some faculty and departmental libraries. Please see www.lib.cam.ac.uk/ for further information.

iDiscover is the University Library's catalogue system. This enables students to simultaneously search across the print items held in the University's libraries and the numerous electronic resources (ebooks, ejournals and databases) to which the University subscribes, together with the open access items in the University's repository. See <http://idiscover.lib.cam.ac.uk> for further information and to manage your loans, requests, fines and saved search results.

Cambridge University Library participates in the Society of College, National and University Libraries (SCONUL) Vacation Access scheme. This enables undergraduate and postgraduate students following taught courses at the University of Cambridge to have reference use of other higher education libraries free of charge. If you request admittance to another library, you should be prepared to present proof of your current membership of Cambridge University (i.e.

your university card). Please check the SCONUL information for each library you wish to access www.sconul.ac.uk/page/about-sconul

The Medical Library

The Medical Library collections support all aspects of medicine and healthcare. The book collection, over 50,000 volumes, contains core textbooks and specialty titles many of which are available electronically and in print. The journal collection is increasingly weighted towards electronic access, with over 104,000 e- titles. The library also facilitates access to the NHS Core Content collection of electronic resources. Please see the [Medical School Library](#) for further information.

The Medical Library provides several computers for student use. These are equipped with common Office applications, the SPSS statistical package, and the EndNote referencing tool. Printing facilities and a scanner are also available. Students can also bring their own devices.

Library training courses, including Medical Library specific courses, can be found at <https://training.cam.ac.uk/cul/event-timetable>

Cambridge Students Union

In July 2020 Cambridge SU replaced the Cambridge University Student Union and the Graduate Union, becoming one central, single union for all students. Each Cambridge MPhil student is automatically a member of the Cambridge SU, which ensures that all students, regardless of level of study, are represented, supported and have their voice heard across the University of Cambridge. The Student Union provides a free independent Advice Service, campaigns for marginalised groups and provides services such as dissertation binding and loan of gowns. Cambridge SU is located at 17 Mill Lane, Cambridge, CB2 1RX. See <https://www.cambridgesu.co.uk/>

Cambridge University Press

You are entitled to 20% discount on books published by Cambridge University Press (CUP) purchased at the Press Bookshop, 1 Trinity Street, Cambridge. You will need to show your university card as evidence of study.

Cambridge University Botanic Garden

You can gain free entry to the University Botanic Garden by presenting your university card.

The University Sports Centre

The University's Sports Centre is in West Cambridge on the outskirts of the city centre. Anyone can become a member and use the facilities and there are a number of competitive membership and pay-as-you-go options. You will need to take your University Card. See www.sport.cam.ac.uk/

Continuation to PhD

The MPhil in Genomic Medicine provides an excellent transition to full-scale PhD research. It introduces and develops research skills and specialist knowledge. The MPhil is also an invaluable introductory experience in preparing and working on a research thesis. There is no automatic progression to a PhD. Those interested in applying for a PhD are encouraged to apply early to the university to ensure they are considered in funding schemes.

A workshop will be held at the end of the Michaelmas term to prepare those applying for PhD interviews. Please let Dr Hearn know during Michaelmas if you would like to take advantage of this session.

Further information about doctoral studies in the University of Cambridge is available on the [Graduate School of Life Sciences'](#) website. This website contains a database of potential projects and supervisors in the Clinical School. You will hear more about the research of members of the Clinical School during the MPhil.

Travel Around Cambridge

Biomedical Campus

There is a bus service (U) that runs frequently during the working day (every 15 minutes) between Madingley Road Park and Ride, University West Cambridge Site and the Biomedical Campus. Prices are discounted to £1 each way for university members. You must show your university card when boarding.

The West Hub

In line with the University's commitment to sustainability, the West Hub does not have any dedicated parking. The nearest public parking is at the Madingley Road Park and Ride, which is just a short walk from the building.

We strongly encourage visitors to use public transport, cycle, or arrive on foot where possible. From the city centre, the West Hub is a:

- 33-minute walk via the scenic Coton footpath
- 10-minute cycle ride
- Short ride on bus routes 4, 8, U, X3, or the Madingley Road Park & Ride service

Cycle parking is available in multiple areas around the building, including the South Plaza off JJ Thomson Avenue. Showers and lockers can be found on the ground floor next to the canteen. Please note, for security reasons, lockers are cleared out each night, so remember to take your belongings with you at the end of the day.

GM1 – Fundamentals of Human Genetics and Genomics

Module Lead – Dr Gemma Chandratillake (Cambridge University Hospitals)

This module contains approximately 30 hours of face-to-face teaching. This introductory module aims to provide the student with an introduction to the key areas of genomics, human genetics and genetic variation. It will prepare participants to understand disease genetics and how genomic medicine can be utilised to elucidate disease mechanisms and biology. In addition, this module will also cover the fundamentals of information governance in the context of genomic medicine and its applications providing underpinning knowledge for later modules in bioinformatics and statistics. This module will serve as a foundation for those wishing to advance their careers within the NHS in genomic medicine. This module provides clear understanding of the structure and variations in genetic material. Covering basic genetics and genomics, it will prepare students to understand the role of genetics in disease and how genomic information can be utilised to elucidate disease mechanisms and biology.

Content

In GM1 you will learn about:

- Structure and function of nucleic acids and chromosomes
- Architecture and organisation of the human genome and genetic variation within it
- DNA sequence variation, type and frequency e.g. single nucleotide variants, small insertions and deletions, copy number variation, rearrangements and tandem repeats
- How variation arises and its extent in populations (e.g. HapMap)
- Gene regulation: enhancers, promoters, transcription factors, silencers
- Epigenetics and imprinting
- Mutational mechanisms: how different types of DNA variants affect gene function or expression to cause disease; correlation of genotype with phenotype
- Concepts of penetrance, expressivity, heterogeneity and pleiotropy
- Modes of inheritance for clinical manifestation of human variation
- Types of genetic testing (prenatal, carrier, predictive, diagnostic, cascade, screening, etc)
- Germline, mitochondrial and somatic mutations
- Chromosomal abnormalities and disease
- Mechanisms of Mendelian disorders
- Interpretation of genetic variation
- Basic Bayesian approach to calculating disease probabilities
- Overview of key ethical, legal and social implications of genomics
- Introduction to some of the principles underlying application of genomic data in clinical practice
- Legislation, Codes of Practice, Caldicott Guardian and Information Commissioner
- Patient identifiable data and information, relationship between data and information
- Information system risks to patient safety, electronic and paper copies, safe havens, encryption, secondary uses of data, audit and research
- Secure information exchange between professionals

- Sharing and communication with patients and careers, consent
- Handling requests for information about patients/clients.

Learning Outcomes

By the end of GM1 you will be able to:

- Discuss the human genome structure and the properties of DNA
- Critique genome architecture and its variation across human populations
- Critically evaluate the regulation of gene expression, transcription and translation
- Appraise and interpret variation in genome structure and sequence in the context of physiological function and disease
- Discuss and analyse epigenetic modifications and imprinting and its role in disease
- Correlate genetic markers to phenotype and interpret output of association studies both for dichotomous and quantitative traits
- Discuss different disease mechanisms and modes of inheritance
- Apply concepts of inheritance and calculate genetic risks for Mendelian conditions
- Discuss and justify the ethical and governance frameworks in place within the NHS and how they apply to medical genomics including patient safety, data sharing and confidentiality
- Identify the range, purposes, benefits and potential risks of sharing, integrating and aggregating clinical data and information.
- Describe and evaluate the purpose, structures, use and storage of health records.



GM2 – Research and Statistical Skills

Module Leads: Dr Tim Hearn (University of Cambridge) and Dr Leonardo Bottolo (University of Cambridge)

This module aims to provide and equip students with the knowledge of statistics and computational tools needed to independently complete genomic medicine research. This module complements the knowledge from the wider course to ensure all students are competent in planning research, and can perform appropriate statistical analysis in R. The module is designed to prepare students for research projects and presents material in this context. This module serves as an informatics foundation for students learning bioinformatics and statistical languages. In particular, the module introduces students to the Genomics England Research Environment, and provides both self-directed, and supervised learning of the Unix command line environment and the R statistics language. This module provides an excellent foundation for performing a research project on the Genomics England 100,000 genomes data, and for students considering undertaking a PhD in future years.

Course content

In GM2 students will learn about:

- How to write a research grant proposal
- Up to date computational and experimental tools for genomic medicine research projects
- The Genomics England research environment and how to perform research on the 100,000 genomes dataset.
- How to use the Unix command line and Galaxy to perform basic genomic analysis
- Fundamental concepts in statistics offering students a foundation and framework for understanding more complex methods.
- Implementation of the statistical methods illustrated during the course using the R environment.
- Basic concepts in using R
- Basic statistical and mathematical concepts including correlation, regression, model selection, and generalized linear models
- Exploration of Bayesian statistics and tools applied for haplotype estimation and Machine Learning tools for non-linear regression will also be explored

Learning Outcomes

By the end of GM2 students will be able to:

- Demonstrate a rounded knowledge of the classic and modern tools and technologies used in research projects. Emphasis will be placed on the overlap between modern and traditional approaches and the type of data they might be presented within a research project
- Understand the conceptual basis for statistical tests, and be aware of the various tools that they can implement when exploring either genomic or expression data
- Choose the correct statistical test based on the data source and experimental design and implement the test in R
- Have a thorough understanding of the types of computational tool available for use in research projects, and the best platform to use them on (Galaxy, command line, HPC etc.)
- Have a good understanding of how to access the central University Linux cluster from an external computer, in preparation for accessing remote High Performance Computing services.



GM3 – Omics Technologies and their Application to Genomic Medicine

Module Leads: Professor Giles Yeo (University of Cambridge) and Dr Brian Lam (University of Cambridge)

This module explores the state-of-the-art genomics techniques used for DNA sequencing (targeted approaches, whole exome and whole genome sequencing) and RNA sequencing, using highly parallel techniques, together with current technologies routinely used to investigate genomic variation in the clinical setting. This module will introduce the bioinformatics approaches required for the analysis of genomic data, which together with data governance covered in GM1 will provide a solid foundation for the Bioinformatics and Statistics modules. The module will also cover the use of array-based methodologies and RNA sequencing in estimating levels of protein expression, micro RNAs and long non-coding RNAs. A comprehensive introduction to metabolomics and proteomics, which are important for the functional interpretation of genomic data and discovery of disease biomarkers will also be included. Students will also learn about the strategies employed to evaluate pathogenicity of variants for clinical reporting.

Teaching of these core technologies and introductory bioinformatics will be facilitated in part by hands-on production of genomic data in which students will take DNA samples through an entire 'omics' workflow.

Content

In GM3 students will learn about:

- Basis of genotyping and detection of genetic variation
- Whole exome and whole genome sequencing, including library preparation methods, sequencing chemistries and platforms
- Hands-on practical experience of an 'omics' workflow from sample to analysed data (virtual for ACY 2020-21)
- Brief overview of methodologies for detecting base substitutions (SNV), small insertions and deletions (indels), copy number variants (CNV) or rearrangements, to include Sanger sequencing, pyrosequencing, ARMS, MLPA, qFPCR, microarray
- Genomic testing strategies as: gene focused, multiple genes, or whole genome or exome, and for detection of sequence, copy number or rearrangements
- RNA expression profiling (expression array) and RNA sequencing
- Metabolomics and proteomics techniques
- Overview of bioinformatics approaches to the analysis of genomic data using Galaxy

Learning Outcomes

By the end of GM3 students will be able to:

- Describe and critically evaluate a range of up-to-date genomic technologies and platforms used to sequence targeted parts of the genome or whole genomes
- Discuss the application of other techniques (for example array comparative genome hybridisation, qPCR) commonly used to interrogate genomic variation in the clinical setting using examples in cancer and rare inherited diseases and infectious diseases
- Acquire the knowledge of selecting appropriate technology platforms for applications in medical genomics either for research or medical diagnostic purposes
- Critique how these techniques and their applications in RNA expression can be applied to metabolomics and proteomic analysis
- Discuss and critically appraise approaches to the bioinformatics analysis and interpretation of 'omics' data
- Critically evaluate the different 'omics' technologies and platforms and their application to genomic medicine and the impact of personalised medicine





GM5 – Bioinformatics, Interpretation and Data Quality Assurance in Genomic Analysis

Module Leads: Dr Tim Hearn (University of Cambridge) and Dr Kenneth Langlands (University of Cambridge)

The aim of this module is to equip students with the skills to analyse and interpret sequencing data, including variant identification, annotation and filtering using appropriate computational and statistical methods. Students will also critically assess the challenges of managing large genomic datasets and the role of databases in variant annotation. This module covers the fundamental principles of bioinformatics with a focus on its impact in both clinical genomics and in research. Students will gain hands-on experience in analysing sequencing data, including quality assessment, alignment to reference genomes, variant calling, and annotation. They will also learn to apply computational skills and statistical methods to handle large genomic datasets and critically evaluate the use of different databases for variant annotation. Through a combination of theoretical sessions and practical exercises, students will develop the essential skills to analyse and interpret genomic data, preparing them to address current and future challenges in managing and utilizing large-scale genomic information in clinical and research settings.

Content

In GM5 students will learn about:

- bioinformatics principles and their application in clinical genomics
- techniques for assessing the quality of raw sequencing data
- common quality metrics (e.g., base quality scores, read length distribution, coverage) and the use of appropriate tools and software
- genome alignment and variant calling:
 - aligning sequencing data to reference genomes using tools like BWA and Bowtie.
 - variant calling methods: single nucleotide polymorphisms (SNPs), insertions and deletions (indels)
 - best practice in variant calling and filtering strategies to identify clinically- relevant variants
- variant annotation and interpretation:
 - annotating variants using databases like dbSNP, ClinVar, and Ensembl.
 - tools for variant annotation
 - interpretation of annotated variants and their clinical significance.
- computational and statistical methods in genomic data analysis:
 - basic computational skills for handling large datasets (e.g., command-line tools, scripting).
 - statistical methods for variant analysis and interpretation.
 - data normalization, variant frequency calculations and significance testing
- databases and resources for genomic data:
 - overview of major genomic databases and their use in variant annotation (e.g., 1000 Genomes, ExAC, gnomAD)

- evaluating the reliability and relevance of different databases
- integrating data from multiple sources for comprehensive variant interpretation.
- challenges in managing large genomic datasets:
 - current and future challenges in genomic data management (e.g., storage, processing power, data security).
 - techniques for efficient data handling and processing (e.g., cloud computing, parallel processing).
 - strategies for maintaining data integrity and privacy in large-scale genomic research.
 - ethical, legal, and social issues in the use of genomic data.
- future trends in genomic data analysis:
 - emerging technologies and methodologies in genomic data analysis.
 - the role of artificial intelligence and machine learning in variant interpretation.

Learning Outcomes

By the end of GM5 students will be able to:

- analyse the quality of sequencing data, align to a reference genome, call and annotate variants, and apply filtering strategies to identify variants in sequencing data
- appraise the use of different databases in variant annotation
- apply relevant basic computational skills and statistical methods to handle and analyse genomic data.
- examine the current and future challenges of managing and manipulating large genomic data sets



GM6 – Application of Genomics in Infectious Disease

Module Leads: Dr Zachary Arden (Wellcome Sanger Institute) and Dr Theo Gouliouris (University of Cambridge)

From this module the student will understand how genomics can be used to better understand the nature, diversity and mechanisms of evolution of microbes causing human disease, provide more accurate diagnosis, predict which drugs are likely to be more effective and monitor treatment and control of infectious disease in individuals and populations.

The student will learn about the genomic structure of infectious agents, implication of acquisition or loss of nucleotides, genes and plasmids on pathogenicity, sensitivity of a pathogen to drug treatment and response to the host. Students will also be taught about current typing approaches for diagnostics and epidemiology of infectious agents, the surveillance systems operating in the UK and abroad and how genome-based approaches can complement/replace these existing methodologies both locally and internationally.

Content

In GM6 students will learn about:

- Infection as a cause of national and global morbidity and mortality
- Transmission of human infections: person to person, food and waterborne, sexually transmitted, vector-borne, zoonosis
- Prokaryotes, their genome, replication and population genetics
- Genomic characterisation of viruses: DNA and RNA genomes, single-stranded, double stranded, segmented
- Genomic comparisons of microbial strains in the context of outbreaks and transmissions in hospitals and the community
- Anti-infective drug action
- Mutation rate and drug resistance
- Genomic evidence of individual susceptibility to specific infection
- Methods and systems for typing and tracking infectious disease locally and internationally.
- Role of genomics in: infectious disease diagnosis, prognosis, drug selection, resistance, monitoring, epidemic control, drug research

Learning Outcomes

By the end of GM6 students will be able to:

- Explain the differences between viral, prokaryote and eukaryote genomes
- Discuss and appraise how the genome sequence of pathogens can be used to track cross infection and outbreaks of infections among the population
- Critically evaluate the emerging action of drugs in controlling infection e.g. Shigella, E-coli, HIV, TB
- Critically evaluate the molecular basis of organism drug resistance in some infections and how this directs drug research
- Evaluate how sequencing of the genome of infective organisms can be used in infectious disease for assessing: diagnosis, sub-classification and strain identity, pathogenicity, drug resistance and drug selection; and for epidemic control
- Explain how genomics facilitates drug research and vaccine design



GM7 – Molecular Pathology of Cancer and Application of Genomics in Cancer

Module Leads: Professor Marc Tischkowitz (University of Cambridge) and Professor George Vassiliou (University of Cambridge)

This module aims to equip the student with detailed knowledge and understanding of the molecular mechanisms involved in cancer development. This will include the ways in which interrogation of a person's own genome and the genome of tumour cells can facilitate the diagnosis and treatment of cancer. This module covers the molecular mechanisms that underlie cancer development, growth and metastasis, and the differences between different cancers. It will explore the different molecular and cellular actions of anti-cancer treatments, the genomic factors affecting response and resistance to treatment, and the research approaches to anti-cancer drug design and development. Broad situations which confer a high cancer risk to a person and/or to other members of the same family will be discussed in the context of how genomic information may be integrated into cancer screening programmes.

Content

In GM7 students will learn about:

- Tumour classification systems
- Cellular properties of tumours: growth, division, invasion, aberrant hormone or toxin production, immunogenicity
- Molecular and functional classification of somatic DNA mutations
- Clonal evolution and subclonal architecture of cancer
- Factors in tumour formation: molecular mechanisms and role of microenvironment, mutagens, mutational processes, molecular signatures & changing classification
- Diagnosis, molecular sub-classification, aggressiveness (prognosis) characterisation of metastases
- Monitoring disease following treatment (medical, surgical or bone marrow transplant)
- Genomic testing of cell free tumour DNA in blood, for diagnosis and monitoring of solid cancers
- Importance of sample quality for tumour genomic analysis
- Understanding the molecular effects of single and the synergy between multiple oncogenic mutations
- Computational/internet resources on Cancer Genomics (COSMIC, mutational calling algorithms, genome browsers, medical/scientific literature)
- The basis of heritable predisposition to cancer
- The role of environmental factors, microorganisms and lifestyle choices on cancer predisposition and protection
- Genomic and cellular markers and choice of treatment regimes in haematological cancer and in solid tumours
- Companion diagnostics in cancer
- Breakthrough tumour, relapse, metastasis and molecular mechanisms

Learning Outcomes

By the end of GM7 students will be able to:

- Understand the mutational processes driving cancer and how these manifest themselves as mutational signatures in cancer genomes
- Appreciate the concepts driving the clonal evolution of different cancers and its parallels to Darwinian natural selection
- Understand the concepts of driver and passenger mutations, as well as the basis of acquired resistance to cancer therapies
- Apply the principles of cancer development and emerging changes in classification
- Compare and contrast the genomic basis of cancer predisposition, and how this is used to identify people and families at higher risk of cancer
- Critically evaluate the impact of genomic advances for the diagnosis, classification, treatment selection and monitoring of cancer (e.g. leukaemia, breast, melanoma, lung cancers)
- Analyse how information from exome and whole genome analysis of tumour tissue can be used to investigate the molecular and cellular processes leading to cancer and inform strategies for drug development and personalised cancer treatments



GM8 – Pharmacogenomics and Stratified Healthcare

Module Leads – Professor Richard Sandford (University of Cambridge), Dr James Whitworth (University of Cambridge) and Dr Emma Hodson (University of Cambridge)

Pharmacogenomics is playing a key role in our health care system. Pharmacogenomics and stratified health care ensure that healthcare professionals tailor the 'right treatment, for the right person, at the right time' and is a fast developing area. This module provides a comprehensive overview of the analytical strategies and techniques used in pharmacogenomics and explore some of the challenges and limitations in this field (availability of patient material for studies of adverse drug reactions which tend to be rare, allelic heterogeneity between different ethnic groups, patient compliance etc.). Biomarkers are the predictive tools for optimising drug response and preventing adverse drug reactions thus this module will also provide an overview of the different type of genomic biomarkers currently in use or emerging. This module describes the complexity of pharmacogenomics and the effect of medication on individuals based on their genetic make-up i.e. tailoring drug treatment to improve patient response and techniques to stratify patients at risk of adverse drug reactions. The module uses examples of known validated pharmacogenomic tests relevant to the use of drug treatments and also use the expertise provided by the major clinical- and academic-industrial research cluster in biomedical science that is developing locally. The module provides a focus to develop interactions with industrial partners and academic groups developing research programs in stratified medicine across a broad range of diseases.

Content

In GM8 students will learn about:

- Genomic basis of: drug reaction, drug efficacy, ethnic differences in both these; and how these are applied in prescribing practice
- Use of genomic information, for targeted drug development
- Companion diagnostics and options for NHS service delivery models
- Different types and examples of genomic-targeted intervention (examples of genomically- targeted clinical, therapeutic or lifestyle choices)
- Genomic biomarkers: SNPs, variability of short sequence repeats, haplotypes, DNA modifications,
- e.g. methylation, deletions or insertions, copy number variants, RNA expression levels, RNA splicing, microRNA levels
- Use of biomarkers in treatments other than cancer.
- The role of academic-industrial partnerships in clinical drug development and clinical trials.

Learning Outcomes

By the end of GM8 students will be able to:

- Discuss and evaluate the mechanism of several examples of genome-determined differential drug response, and drug reaction
- Appraise the strategies and analytical approaches for stratifying patients for optimal drug response or adverse drug reactions including ethnic differences, and how these translate into ‘companion diagnostics’
- Identify and analyse the challenges and limitations of pharmacogenetic studies
- Identify and evaluate the different types of current and emerging biomarkers used in personalised medicine
- Discuss and critically evaluate how genomic information can enable development of drugs targeted for particular genotypes
- Identify the ethical, legal and social issues (ELSI) that could accompany patient stratification for healthcare advice or intervention and defend the use of patient stratification to improve the diagnosis and treatment of disease
- Explain the implication of pharmacogenomic results in current clinical practice considering relevant national guidance.



GM9 – Introduction to Counselling and Communication Skills in Genomic Medicine

Module Leads: Eleanor Davies (Cambridge University Hospitals) and Isobel Turbin (Cambridge University Hospitals)

This is an introduction to counselling skills for genomics. The aim of this module is to equip students with the knowledge, communication and counselling skills and appropriate attitudes and behaviours towards the diagnosis and management of patients whose care will be influenced by genomic investigations.

Students undertaking this module are taught how to communicate and provide appropriate support to individuals and their families. Development of counselling skills is achieved via theoretical and practical sessions. Students are taught to understand the importance of a family history and communication of pathogenic and/or uncertain results. Patient involvement is an integral feature of this module. The patient perspective will also be considered extensively within the role play delivery.

Content

In GM9 students will learn about:

- Communication and counselling skills
- Consent and what it means in relation to the 100,000 Genomes Project
- Ethical and social implications of genomic testing
- How to record and interpret a family history, recognising what is or may be relevant
- How to verify personal and family history information; consent, confidentiality, access to records
- Different purposes of genomic testing
- Approaches to prenatal testing, pre-implantation testing (PGD) and pre-conception carrier screening in relation to new technologies
- Strategies of approach to lifelong patient management of whole genome information
- Managing and explaining complex genome results
- Sources for patient support: patient support groups, on-line resources, other resources

Learning Outcomes

By the end of GM9 students will be able to:

- Explain and justify the importance of and application of informed consent in the field of genomic medicine
- Explain the different purposes of genomic testing in patients with rare inherited diseases, cancer and infectious diseases
- Explain genomic results in terms of diagnosis prediction and uncertainty
- Discuss the concepts of genetic and genomic predispositions to illnesses



- Discuss the consequences of genomic test results on the patient and the wider family including incidental findings drawing on the published evidence base and personal experiences of patients, carers and the wider family
- Explain the range of screening pathways used to test for inherited and acquired disorders and evaluate their effectiveness including how they support clinical decision making
- Discuss and critically evaluate current and potential future ethical, legal and social issues (ELSI) of genome testing and whole genome sequencing
- Demonstrate advanced communication skills required to explain complex genomic concepts to a lay audience
- Demonstrate appropriate empathetic and culturally competent practice in supporting patients through the genomic testing process



GM10 – Epigenetics and Disorders of the Epigenome

Module Lead: Professor Eamonn Maher (Aston University) and Professor Miguel Constância

The completion of the human genome project has paved the way for major advances in our understanding of human genetic disease and the pathogenesis of cancer. Nevertheless, these advances have also revealed that genetics is only part of the story and there is increasing recognition that epigenetics will be critical for understanding human disease and the practice of genomic medicine. Cambridge is the leading European centre for epigenetic research and this module will draw on world-leading local scientists to provide expert coverage of this fascinating topic.

Content

In GM10 students will learn about:

- Definition and history of epigenetics
- Elements of epigenetic gene regulation: chromatin packaging, nucleosomes, histone marks, establishment and maintenance of DNA methylation, non-coding RNAs etc.
- Methodologies for investigating epigenetics regulation in health and disease
- Genomic resources for epigenetic research and interpretation. The Epigenome Project
- Role of model organisms in elucidating epigenetic control mechanisms
- Mechanisms of epigenetic regulation of gene expression in normal development (including nuclear reprogramming, X-chromosome inactivation and genomic imprinting)
- Epigenetics and inherited disease: inherited disorders caused by mutations in genes regulating epigenetic processes, human imprinting disorders, assisted reproductive technologies
- Epigenetics and acquired disease: epigenetics and cancer, epigenetics and gene/environmental interactions, epigenetics and ageing
- Epigenetics and therapeutics

Learning Outcomes

By the end of GM10 students will be able to:

- Understand the role of epigenetic factors in regulating gene expression
- Appreciate the critical role of epigenetics in normal development and human health
- Critically appraise methodologies for epigenetic studies of human disease
- Understand the role of epigenetic abnormalities in human disease, and in particular, the role in inherited disorders and neoplasia

- Describe the relevance of epigenetics for gene-environment interactions and the significance of epigenetics for therapeutic strategies for human diseases such as cancer



GM11 – Variant Interpretation

Module Leads: Dr Tim Hearn (University of Cambridge) and Dr Kenneth Langlands (University of Cambridge)

This module will equip students with the understanding and skills required to carry out genomic variant classification in a clinical setting. The module will guide students through the process of clinical variant interpretation and the basics of relevant bioinformatics pipelines. It will consider the fundamental principles of variant classification including the variety of evidence used (including population, phenotype, inheritance, predictive and functional data). There will be a focus on how this process is carried out in a clinical setting including an exploration of relevant guidelines, multidisciplinary team working and interpreting genomic variant reports

Content

In GM11 students will learn about:

- genomic data flow from the patient, through to the laboratory, to the clinician and then back to patient.
- bioinformatic pipelines including assessment of data quality, alignment, variant identification and annotation processes
- the use and limitations of population databases and case control studies in variant classification
- approaches for assessing the functional effect of variants
- relevant computational and predictive data sources and how to use these in variant classification
- the role of data gathered by the clinical team in variant classifications, i.e. good phenotyping and inheritance data
- use of multiple database sources and clinical literature in variant classification
 - principles of integration of laboratory and clinical information, and location of best practice guidelines for indicating the clinical significance of results (for example, American College of Medical Genetic and Genomics (ACMG), and UK guidelines for bioinformatics and variant classification)
- the multidisciplinary team process, including MDT meetings, in clinical variant interpretation
- ethical and legal issues, including:
 - o Patient data security in the various aspects of variant identification
 - o Variation in population data availability for different ethnicities and the impact on classification for patients
 - o Changes in variant classification over time and how this can impact patient care

Learning Outcomes

By the end of GM11 students will be able to:

- interpret evidence (including population, phenotype, inheritance, predictive and functional data) used for variant classification.
- apply relevant national and international guidelines to classify genomic variants.
- appraise the strengths and limitations of variant interpretation in a multidisciplinary team (MDT) setting.
- evaluate a range of genomic variant reports in human disease.





GM12 – Genomics of Common and Rare Disease

Module Leads: Dr Katherine Schon (University of Cambridge), Dr Ramsay Bowden (University of Cambridge) and Dr Laura Fachal (Wellcome Sanger Institute)

The number of rare monogenic disorders is estimated to be greater than 7000, but only in approximately half of these are the underlying genes known. Common conditions such as intellectual disability, diabetes, and schizophrenia are thought to arise from a complex interplay of genetic and environmental factors but deeper understanding of the genetic and mechanistic basis of these conditions is necessary for clinical translation.

The module offers a comprehensive overview of the traditional and current strategies and techniques used to identify genes responsible for both common multifactorial and rare inherited diseases, focusing mainly on the latter. This module will initially explore the clinical presentation and course of a range of common and rare inherited diseases. The principles and practise of medical genetics, and the management and treatment of probands and their families will be discussed. In addition, the role of genomics in a care pathway will be examined including the patient and family perspective. Building on knowledge gained in GM2, students will further explore the analytical challenges in genomics as applied to rare inherited diseases.

Content

In GM12 students will learn about:

- Clinical presentation and course of a range of rare inherited and common diseases
- Principles and practise of medical genetics; risk stratification and management of patients and their families
- Approaches and techniques used to identify genes responsible for common and rare inherited diseases
(e.g. candidate gene, positional mapping, genome wide association studies, exome/whole genome sequencing, use of population data sets)
- An overview of the techniques used for functional validation of putative disease-associated mutations
- Basic statistics to aid interpretation of Genome Wide Associated Studies (GWAS) and analysis of populations
- The Genomics England 100,000 Genomes Project and data infrastructure
- Selection of tractable cases with unmet diagnostic need suitable for whole genome analysis
- Analytical challenges in genomics as applied to rare inherited diseases including; the potential of electronic health records to enrich patient data, the importance of phenotyping and use of databases such as ClinVar, the use of large population datasets and sharing information e.g. Human Variome Project
- Impact of patient on-line access to their health records, test results etc. on medical genomics.

Learning Outcomes

By the end of GM12 students will be able to:

- Examine the landscape of common and rare inherited diseases
- Explain the genetic architecture of common and rare inherited diseases
- Critically evaluate traditional and current approaches used to identify genetic variation in common and rare inherited diseases
- Synthesise information gained from exome/whole genome analysis with patient information / medical records to determine diagnosis, penetrance or prognosis for a number of examples of common and rare inherited conditions
- Discuss and evaluate the Genomics England Programme and the Data Infrastructure
- Identify phenotype, select cases and relevant family information for whole exome or whole genome-based approaches for hypothesis free whole exome or whole genome sequencing
- Discuss and critically evaluate the implications of patient access to their medical records and clinical information for medical genomics, inter- professional practice and multidisciplinary care
- Demonstrate understanding of the range of methodologies that can be used to functionally validate putative disease mutations.





GM13 – Genomics of Obesity and Metabolic Disease

Module Leads: Dr Robert Legg (University of Cambridge) and Dr Kenneth Langlands (University of Cambridge)

As our understanding of the contribution of genetics to a range of human disorders grows, so effective screening strategies reveal new insights that will enhance health outcomes. This module will enable you to develop your understanding of the increasing role of genomics in the diagnosis and management obesity and metabolic disease (along with their sequelae) while encouraging you to integrate ideas from across the curriculum. You will learn about common and rare genetic contributors to obesity and metabolic disease, and how this informs decision making by multidisciplinary teams. Equitable access to high-quality care and the impact of living with obesity and metabolic disease will be explored. You will gain practical experience of how genome-wide association studies are performed and the role of polygenic scoring in disease stratification. The introduction of drugs such as GLP-1 agonists has transformed obesity care, and you will learn of the next generation of drugs in development.

Content

In GM13 students will learn about:

- Molecular mechanisms underpinning obesity and metabolic disease
- Impact on lipid metabolism and cardiovascular sequelae
- Common clinical presentations and management
- Equity in access to care and the impact of drugs such as GLP-1 agonists
- Patient and family perspective on living with obesity and metabolic disease
- Environmental and socio-economic determinants of health
- The role of MDTs in clinical decision making
- Screening strategies
- The use of genome wide-association studies to identify common and rare contributors to obesity and metabolic disease, and the role of polygenic risk scoring in disease stratification
- Emerging genomic technologies and innovation in drug design

Learning Outcomes

By the end of GM13 students will be able to:

- Explain mechanisms in monogenic and polygenic causes of disease
- Evaluate current approaches to the diagnosis and management of obesity and metabolic disease
- Critically evaluate the contribution of genetics to disease outcomes
- Explain how phenotypic data is integrated with genomic information in diagnostic and treatment pathways
- Analyse large-scale genomic data, interpret variants and generate clinically-relevant insights



- Explain and contrast the contributions of the members of multi-disciplinary teams
- Discuss the contribution of environmental and socio-economic factors to obesity and metabolic disease and evaluate strategies to reduce health inequalities
- Appraise emerging therapeutic targets

